

**MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL**

Paper Code : BS-M401/M(CS) 401/M(CS)401 Numerical Methods(BS)

UPID : 004401

Time Allotted : 3 Hours

Full Marks :70

*The Figures in the margin indicate full marks.**Candidate are required to give their answers in their own words as far as practicable***Group-A (Very Short Answer Type Question)**

1. Answer any ten of the following :

[1 x 10 = 10]

- (I) What is the restriction of the arguments on the interpolating points for Lagrange interpolation formula ?
- (II) What is the degree of precision of Trapezoidal rule ?
- (III) What is the convergence rate of Regula-falsi method?
- (IV) What is the order of truncation error in Runge-Kutta Method 4-th order?
- (V) What is the approximate number of 3.4506531 when round off to 4 places of decimal ?
- (VI) What is the relation between backward difference operator and shift operator?
- (VII) What is the degree of the interpolation polynomial of a function whose values are known at 8 points?
- (VIII) In simpson's 1/3 rd rule for finding $\int_a^b f(x)dx$, $f(x)$ is approximated by

(a)Line segment (b) parabola (c) circular sector (d) part of ellipse

- (IX) For which condition of the pivotal elements, the Gauss elimination method fails?
- (X) Why we use Runge-Kutta method?
- (XI) What is the number of significant digits in 12340?
- (XII) What is the value of 2nd order forward difference of 1st degree polynomial?

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

- 2. Find the smallest positive root of the equation $3x^3 - 9x^2 + 8 = 0$ correct up to four decimal places, using Newton-Rapson method. [5]
- 3. Find the absolute, relative and percentage error if $1/3$ is approximated by 0.333. [5]
If $5/6$ is represented by the approximate number .8333, compute absolute, relative, percentage errors.
- 4. Find the approximate value of $I = \int_0^1 \frac{dx}{1+x}$, when the interval is (0,1) and $h = \frac{1}{4}$ by using simpson's 1/3 rd rule. [5]
- 5. By constructing a difference table, find the sixth term of the series 8,12,19,29,42..... [5]
- 6. Using Newton -Raphson method , find a real root of the equation $x^4 - x - 1 = 0$, correct to three decimal places. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

- 7. Find the Absolute ,Relative and Percentage errors when 5.0214 is rounded off to 3 significant figures. [15]
If $5/3$ is approximated to 1.667, find the Absolute, Relative and Percentage errors.
Find the percentage error when $7/6$ is approximated by 1.16.
- 8. (a) Evaluate $\int_0^2 \frac{x}{\sin x} dx$, where the interval is (0, 2) by using Trapezoidal rule taking $n = 8$. [8+7]

(b) Estimate the value of the following integral by Simpson's $\frac{1}{3}$ rule taking 4 strips:

$$\int_1^3 \frac{1}{x} dx.$$

9. (a) Evaluate the error terms of Trapezoidal rule. [15]

(b) A solid of revolution is formed by rotating about the x-axis, the area between the x-axis, the lines $x=0, x=1$ and a curve through the points with the following coordinates

x	0	.25	.50	.75	1
y	1	.9896	.9589	.9089	.8415

Find the volume of the solid of revolution.

10. (a) Solve the following system of equations by using Gauss-seidel method [15]

$$10x - y - z = 13$$

$$x - 10y + z = 36$$

$$x + y - 10z = -35.$$

(b) Find the inverse of the matrix

$$\begin{pmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{pmatrix}$$

Solve the following system of equations by Gauss-elimination method

$$5x - y - z = 3$$

$$-x + 10y - 2z = 7$$

$$-x - y + 10z = 8.$$

11. [8+7]

Find the smallest positive root of the equation $e^x = 4 \sin x$ correct to four decimal places by Bisection method

Find $\sqrt[4]{13}$ by Newton-Raphson method

*** END OF PAPER ***